

CODECHECK certificate 2026-002

<https://codecheck.org.uk/register/certs/2026-002/>



Table 1: CODECHECK summary

Item	Value
Title	<i>Preserving privacy of spatial distances using randomized geometric surface calculations</i>
Author(s)	Jonas Klingwort  , Sarah Redlich
Reference	https://doi.org/10.5311/JOSIS.2025.31.375
Repository	https://github.com/jkwort/ppdc
Codechecker(s)	Linus Dexter Hackel 
Date of check	2026-06-29
Summary	TODO add summary

Table 2: Summary of output files generated

File	Comment	Size (b)
<code>figure_1_step_1.pdf</code>	Figure 1 - Step 1 - Generate base map with Germany and bounding box	263198
<code>figure_1_step_2.pdf</code>	Figure 1 - Step 2 - Calculate true distances	263388
<code>figure_1_step_3.pdf</code>	Figure 1 - Step 3 - Draw five random coordinates and obtain triangles	263972
<code>table_1.csv</code>	Table 1 - Calculate surface areas of the triangles	226
<code>table_2.csv</code>	Table 2 - Distribution of geographical distances (Haversine distance in km) for Germany and the Netherlands.	176

Summary

TODO add summary

CODECHECKER notes

For a majority of the scripts included in the `code` directory of the [code repository](#), the data first had to be created with the script: `03_section_6_prep_evaluation.R` which is provided in the `code` directory as well. The problem is that the authors used a large CPU HPC cluster to calculate these datasets, which I as a codechecker don't have access to. But since the datasets are created in the loop beginning in line 79, this loop can also be stopped earlier to produce a portion of the needed base datasets. After

consulting with [Daniel Nüst](#), we decided that the portions of the base datasets would still be sufficient to check the reproducibility of the code.

When reproducing Table 2, I noticed that the values are all equal to the paper, but the Header of the Table isn't matching. This could either be down to me using the reduced base datasets for reproduction, or the authors later renaming the Column Headers in their paper. My column headers are: **Country, Min., 1st Quantile, Median, Mean, 3rd Quantile, Maximum** , while the papers column headers are: **Country, Min., 25th Quartile, 50th Quartile, Mean, 75th Quartile, Max.** .

Recommendations to the authors

TODO

Manifest files

CSV files

table_1.csv

Author comment: *Table 1 - Calculate surface areas of the triangles*

R_n	base_side_of_triangle	height_of_the_base_side	surface_area
1	1.36115	1.75236	1.19261
2	1.36115	5.52275	3.75864
3	1.36115	2.72697	1.85591
4	1.36115	3.65482	2.48738
5	1.36115	2.67144	1.81811

table_2.csv

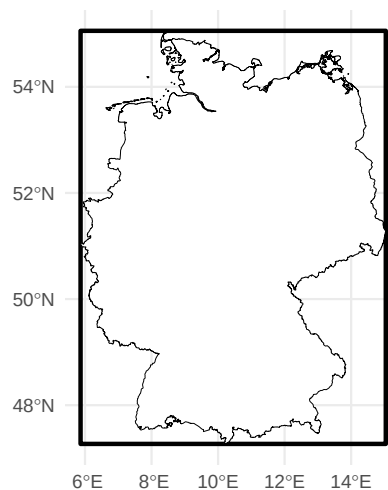
Author comment: *Table 2 - Distribution of geographical distances (Haversine distance in km) for Germany and the Netherlands.*

Country	Min.	1 st Quantile	Median	Mean	3 rd Quantile	Maximum
Germany	0.11	199.62	315.03	320.39	431.35	875.67
Netherlands	0.01	53.9	88.03	95.35	131.24	323.88

Figures

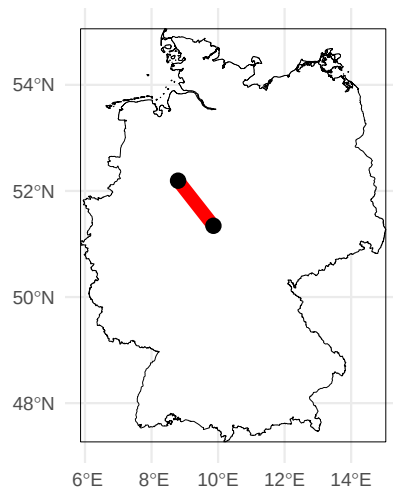
figure_1_step_1.pdf

Author comment: *Figure 1 - Step 1 - Generate base map with Germany and bounding box*



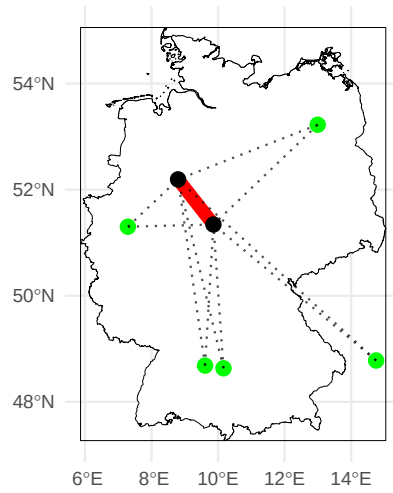
figure_1_step_2.pdf

Author comment: *Figure 1 - Step 2 - Calculate true distances*



figure_1_step_3.pdf

Author comment: *Figure 1 - Step 3 - Draw five random coordinates and obtain triangles*



Acknowledgements

TODO: acknowledge the authors who helped in reproducing the results and figures of the paper.

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Citing this document

Linus Dexter Hackel (2026). CODECHECK Certificate 2026-002. Zenodo. <https://codecheck.org.uk/register/certs/2026-002/>

About CODECHECK

This certificate confirms that the codechecker could independently reproduce the results of a computational analysis given the data and code from a third party. A CODECHECK does not check whether the original computation analysis is correct. However, as all materials required for the reproduction are freely available by following the links in this document, the reader can then study for themselves the code and data.

About this document

This document was created using [codecheck-py](#) (a Python-base template for creating [CODECHECK](#) certificates). The CODECHECK details are filled into a [jupyter notebook](#) which is then converted into Markdown via [nbconvert](#). Afterwards it gets converted into [Typst](#) using [cmarker](#) and then into PDF using Typst. `sh notebook_to_pdf.sh` will regenerate the report file.

```
import session_info2 as si
si.session_info(os=True, cpu=True, gpu=True, dependencies=True)
```

```
platformdirs 4.5.1
comm 0.2.3
charset-normalizer 3.4.4
pure_eval 0.2.3
PyYAML 6.0.3
stack_data 0.6.3
debugpy 1.8.17
numpy 2.3.5
jedi 0.19.2
session-info2 0.3
six 1.17.0
executing 2.2.1
python-dateutil 2.9.0.post0
parso 0.8.5
pytz 2025.2
ipykernel 7.1.0
jupyter_client 8.7.0
packaging 25.0
idna 3.11
prompt_toolkit 3.0.52
tornado 6.5.3
decorator 5.2.1
setuptools 80.9.0
psutil 7.1.3
asttokens 3.0.1
ipython 9.8.0
pyzmq 27.1.0
urllib3 2.6.1
requests 2.32.5
traitlets 5.14.3
Pygments 2.19.2
jupyter_core 5.9.1
Brotli 1.2.0
```

```
PySocks 1.7.1
tabulate 0.9.0
certifi 2026.1.4 (2026.01.04)
pandas 3.0.0
wcwidth 0.2.14
colorama 0.4.6
----
Python 3.14.2 | packaged by conda-forge | (main, Dec 6 2025, 11:21:58) [GCC 14.3.0]
OS Linux-6.18.33.2-microsoft-standard-WSL2-x86_64-with-glibc2.35
CPU 16 logical CPU cores, x86_64
GPU No GPU found
Updated 2026-06-29 14:11
```

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